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Thursday, April 23, 2020   10:53 PM

Vector Spaces / subspaces

$$A \cdot x = \begin{bmatrix} 5 & 1 & -4 & -2 & 1 \\ 0 & 2 & 4 & -3 & -1 \end{bmatrix} \begin{bmatrix} -3 \\ 4 \\ 1 \\ 5 \\ 2 \end{bmatrix}$$

$$= \begin{bmatrix} -15 + 4 - 4 - 10 + 2 \\ 0 + 8 + 4 - 15 - 2 \end{bmatrix}$$

$$= \begin{bmatrix} -23 \\ -5 \end{bmatrix} \in \mathbb{R}^{2\times 1} \equiv \mathbb{R}^2$$

① 判断是不是 subspaces.

a) $C := \{(x,y) \in \mathbb{R}^2,\ x^2+y^2=1\}$

C is the unit circle in $\mathbb{R}^2$.

it's not subspace because $(0,0) \notin C$.

$(0^2+0^2 \neq 1)$

b) $P := \{(x,y) \in \mathbb{R}^2 \mid x \geq 0 \text{ and } y \geq 0\}$

$(0,0) \in P$, so P can be vector space.

Is it closed under addition?

If $(x,y) \in P$.

$(u,v) \in P$

So $\left.\begin{matrix} x \geq 0,\ y \geq 0 \\ u \geq 0,\ v \geq 0 \end{matrix}\right\} \begin{matrix} x+y \geq 0 \\ u+v \geq 0 \end{matrix}$

$(x,y) + (u,v) \in P$ 正确.

But. if $(x,y) \in P$ with $x, y > 0$

for example

$\underset{\lambda}{\underset{\parallel}{(-1)}} \cdot (x,y) = (-x,-y) \notin P$

Ex: $(x,y) = (1,2) \in P$

but $(-1) \cdot (1,2) = (-1,-2) \notin P$

So P is not a subspace.

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② Are the following sets Subspace of $\mathbb{R}^3$?

a) $S_1 = \{(x,y,z) \in \mathbb{R}^3 \mid x^2+y^2+z^2=1\}$

It's not a subspace because $(0,0,0) \notin S_1$

$(0^2+0^2+0^2 = 0 \neq 1)$

b) $S_2 = \{(x,y,z) \in \mathbb{R}^3 \mid x \geq 0, y \geq 0, z \geq 0\}$

Is not a subspace.

because $(1,2,1) \in S_2$ but

$(-1) \cdot (1,2,1) = (-1,-2,-1)$

c) $S_3 = \{(x,y,z) \in \mathbb{R}^3 \mid 2x-3y+z=0\}$

$(0,0,0) \in S_3$

$2 \cdot 0 - 3 \cdot 0 + 0 = 0$

is a plane that goes through $(0,0,0)$

and locates the natural vector $(2,-3,-1)$

Is it a subspace?

We have to check that if $a, b \in S_3$

$\Rightarrow a+b \overset{?}{\in} S_3$.

So, $a = \begin{pmatrix} x \\ y \\ z \end{pmatrix} \in S_3$, $b = \begin{pmatrix} u \\ v \\ w \end{pmatrix} \in S_3$

$2x-3y+z=0 \qquad 2u-3v+w=0$

What is $a+b$

$$a+b = \begin{pmatrix} x+u \\ y+v \\ z+w \end{pmatrix}$$

We evaluate.

$2(x+u) - 3(y+v) + (z+w)$

$= 2x + 2u - 3y - 3v + z + w$

$= (2x-3y+z) + (2u-3v+w)$

$= (\Delta)$

$$a+b = \begin{pmatrix} x+u \\ y+v \\ z+w \end{pmatrix} \in S_3$$

AND if $\lambda \in \mathbb{R}$ any number

and $a \in S_3 \Rightarrow \lambda a \overset{?}{\in} S_3$ su

$$\lambda \cdot a = \lambda \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} \lambda x \\ \lambda y \\ \lambda z \end{pmatrix} \in S_3$$

$\Leftrightarrow$

$2(\lambda x) - 3(\lambda y) + 1 \cdot (\lambda z) = 0$

$\lambda \underbrace{(2x-3y+z)}_{=0 \ (a \in S_3)} = 0$ ✓

So S_3 is a subspace in $\mathbb{R}^3$.

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d) $S_4 := \{(x,y,z) \in \mathbb{R}^3 \mid 2x-3y+z=5\}$

S_4 is not a subspace,

since $(0,0,0) \notin S_4$.

$2 \cdot 0 - 3 \cdot 0 + 0 = 0 \neq 5$.

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e) $S_5 = \{(x-y, 3x, 2x+y) \in \mathbb{R}^3 ; x,y \in \mathbb{R}\} \subseteq \mathbb{R}^3$

So if $a, b \in S_5 \Rightarrow$

$a = \begin{pmatrix} x-y \\ 3x \\ 2x+y \end{pmatrix}$, with some $x, y \in \mathbb{R}$.

$b = \begin{pmatrix} u-v \\ 3u \\ 2u+v \end{pmatrix}$, with some $u, v \in \mathbb{R}$

$$\Rightarrow a+b = \begin{pmatrix} x-y \\ 3x \\ 2x+y \end{pmatrix} + \begin{pmatrix} u-v \\ 3u \\ 2u+v \end{pmatrix}$$

$$= \begin{pmatrix} (x+u)-(y+v) \\ 3(x+u) \\ 2(x+u)+(y+v) \end{pmatrix} = \begin{pmatrix} \alpha-\beta \\ 3\alpha \\ 2\alpha+\beta \end{pmatrix}$$

Where:

$\alpha = x+u \in \mathbb{R}$

$\beta = y+v \in \mathbb{R}$

$\Rightarrow a+b \in S_5$ ✓

Similarity if: $\lambda \in \mathbb{R}$

$a = \begin{pmatrix} x-y \\ 3x \\ 2x+y \end{pmatrix} \in S_5$ with $x, y \in \mathbb{R}$

Then $\lambda \cdot a = \begin{pmatrix} \lambda(x-y) \\ \lambda 3x \\ \lambda(2x+y) \end{pmatrix}$

$$= \begin{pmatrix} \lambda x - \lambda y \\ 3\lambda x \\ 2(\lambda x) + \lambda y \end{pmatrix} = \underset{\overset{\uparrow}{S_5}}{\begin{pmatrix} \alpha-\beta \\ 3\alpha \\ 2\alpha+\beta \end{pmatrix}}$$

where $\alpha = \lambda x \in \mathbb{R}$

$\beta = \lambda y \in \mathbb{R}$

$\Rightarrow \lambda a \in S_5$ So S_5 is a subspace in $\mathbb{R}^3$.